RACE
Regular Analysis Through Continuous Exercise

Chemical Equilibrium : RACE-01

Introduction: Equilibrium in Chemical Processes - Dynamic Equilibrium ; Law of Mass Action

1. 48 g of SO_3, 12.8 g of SO_2 and 9.6 g of O_2 are present in one litre. The respective active masses will be
(A) 1.0, 0.5 and 0.3 (B) 0.6, 0.2 and 0.3 (C) 0.6, 0.4 and 0.2 (D) 1.0, 0.5 and 1.5

CCE183

Law of Chemical Equilibrium and Equilibrium constant; Kp

2. For the reaction $C(s) + CO_{2(g)} \rightleftharpoons 2CO_{(g)}$ the partial pressure of CO and CO_2 are 2.0 and 4.0 atm. respectively at equilibrium. The KP for the reaction is
(A) 0.5 (B) 4.0 (C) 8.0 (D) 1

CCE184

3. The equilibrium constant (K_P) for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 16. If the volume of the container is reduced to one-half its original volume, the value of K_P for the reaction at the same temperature will be :-
(A) 32 (B) 64 (C) 16 (D) 4

CCE150

4. If the forward rate constant of reversible reaction is 0.16 and backward rate constant is 4×10^4, then the equilibrium constant will be.
(A) 2.25×10^{-6} (B) 2.5×10^{-5} (C) 4×10^{-6} (D) 4×10^{-5}

CCE185

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Equilibrium constant; Kc

5. At 1000 K, the value of K_p for the reaction:

$A(g) + 2B(g) \rightleftharpoons 3C(g) + D(g)$

is 0.05 atm the value of K_c in terms of R would be
(A) 20000 R (B) 0.02 R (C) 5×10^{-5} R (D) $5 \times 10^{-5} \times R^{-1}$

CCE186

6. Match List –I (hypothetical reactions) with List–II (ratio $\frac{K_P}{K_C}$ for the given reactions) and select the correrct answer using the code given below the lists :-

(a) $A_2(g) + 3B_2(g) \rightleftharpoons 2AB_3(g)$ (i) $(RT)^{-2}$

(b) $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$ (ii) $(RT)^0$

(c) $A(s) + 1.5\ B_2(g) \rightleftharpoons AB_3(g)$ (iii) $(RT)^{\frac{1}{2}}$

(d) $AB_2(g) \rightleftharpoons AB(g) + 0.5\ B_2(g)$ (iv) $(RT)^{-\frac{1}{2}}$

CODES :

	a	b	c	d
(A)	(i)	(ii)	(iii)	(iv)
(B)	(ii)	(i)	(iv)	(iii)
(C)	(i)	(iv)	(iii)	(ii)
(D)	(i)	(ii)	(iv)	(iii)

CCE149

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Application of Equilibrium Constant

7. Given that the equilibrium constant for the reaction

$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

has a value of 278 at a particular temperature. What is the value of the equilibrium constant for the following reaction at the same temperature ?

$SO_{3(g)} \rightleftharpoons SO_{2(g)} + \frac{1}{2}O_{2(g)}$

(A) 6.0×10^{-2} (B) 1.3×10^{-5}

(C) 1.8×10^{-3} (D) 3.6×10^{-3}

CCE187

8. K_P value for

$C_2H_4 + H_2 \rightleftharpoons C_2H_6$ is 5×10^{18} atm^{-1} and K_P for

$C_2H_2 + H_2 \rightleftharpoons C_2H_4$ is 5×10^{26} atm^{-1} at 50°C.

Calculate KP for the reaction

$2H_2 + C_2H_2 \rightleftharpoons C_2H_6$

(A) 2.5×10^{45} (B) 21×10^{8}

(C) 1.5×10^{-45} (D) 2.25×10^{-45}

CCE188

9. If the value of an equilibrium constant for a particular reaction is 1.6×10^{12}, then at equilibrium the system will contain :-

(A) mostly reactants

(B) mostly products

(C) similar amounts of reactants and products

(D) Only reactants

CCE151

Degree of Dissociation

10. If 0.5 mole H_2 is reacted with 0.5 mole I_2 in a ten-litre container at 444° C and at same temperature value of equilibrium constant Kc for the reaction: $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$, is 49, the ratio of [HI] and $[I_2]$ will be:-

(A) 7 (B) $\frac{1}{7}$

(C) $\sqrt{\frac{1}{7}}$ (D) 49

CCE189

11. In the reaction A+ 3B $\rightleftharpoons$ 2C, initially 10 moles each of A and B were taken. If equilibrium pressure is 8 atm and dissociation of A is 20% then K_P in $(atm)^{-2}$ will be :-

(A) $\frac{1}{4}$ (B) $\frac{1}{8}$ (C) $\frac{1}{16}$ (D) $\frac{1}{2}$

CCE190

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12. $2ICl_{(g)} \rightleftharpoons I_{2(g)} + Cl_{2(g)}$

Equilibrium constant Kc for given reaction is 0.14

If initally 0.6M ICl was taken then what will equilibrium concentration of l_2 is :-

(A) 0.1275 M (B) 0.3250 M

(C) 0.35 M (D) 0.2550 M

CCE191

13. For a reaction $A_{(g)} \rightleftharpoons B_{(g)}$ if initial pressure at 300°C is 1 atm and its K_P is 4, Then calculate P_B at equilibrium state.

(A) $\frac{1}{5}$atm (B) $\frac{2}{5}$atm

(C) $\frac{4}{5}$atm (D) $\frac{3}{5}$atm

CCE147

14. In a container 96g O_2 and 335g Ne are present. If total pressure of gaseous mixture is 25 atm. Then what will be partial pressure of Ne?

(A) 21.20 atm (B) 15.7 atm

(C) 42.6 atm (D) 3.56 atm

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CCE148

15. Evaluate K_P for the reaction : $H_2 + I_2 \rightleftharpoons 2HI$. If 2 moles each of H_2 and I_2 are taken initially. The equilibrium moles of HI are 2.

(A) 2.5 (B) 4

(C) 0.25 (D) 1.0

CCE152

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	C	C	D	D	A	A	B	A
Question	11	12	13	14	15					
Answer	B	A	C	A	B					

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RACE Regular Analysis Through Continuous Exercise

Chemical Equilibrium : RACE-02

Degree of Dissociation

1. AB dissociates as $2AB\,(g) \rightleftharpoons 2A\,(g) + B_2\,(g)$

When the initial pressure of AB is 500 mm, the total pressure becomes 625 mm when the equilibrium is attained. Calculate K_P for the reaction assuming volume remains constant.

(A) 500 (B) 125 (C) 750 (D) 375

CCE153

2. At 60° C and initial pressure of 1atm of N_2O_4 is 50% dissociated into NO_2 then Kp is :-

(A) 1.33 atm (B) 2 atm (C) 2.67 atm (D) 3 atm

CCE154

3. At same temperature N_2O_4 is dissociated to 40% & 50% at total pressure P_1 & P_2 atm respectively in NO_2. Then the ratioof P_1 & P_2 is

(A) $\frac{4}{5}$ (B) $\frac{7}{4}$ (C) $\frac{4}{7}$ (D) None of these

CCE155

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4. For the dissociation reaction

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$,

the degree of dissociation (α) in terms of K_p and total equilibrium pressure P is

(A) $\alpha = \sqrt{\frac{4P+K_p}{K_P}}$ (B) $\alpha = \sqrt{\frac{K_p}{4P+K_P}}$

(C) $\alpha = \sqrt{\frac{K_p}{4P}}$ (D) None of these

CCE156

5. What per cent of CO_2 in air is just sufficient to prevent loss in weight when $CaCO_3$ is heated at 100°C.

Equilibrium constant K_p for $CaCO_3\,(s) \rightleftharpoons CaO\,(s) + CO_2\,(g)$ is 0.0095 atm at 100°C.

(A) 0.95 % (B) 0.29 % (C) 0.05 % (D) 0.71 %

CCE158

6. A 20 litre container at 400 K contains $CO_2(g)$ at pressure 0.4 atm and an excess of SrO (neglect the volume of solid SrO). The volume of the container is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be :-

(Given that: $SrCO_3(s) \rightleftharpoons SrO(s) + CO_2(g)$, Kp = 1.6atm)

(A) 10 litre (B) 4 litre (C) 2 litre (D) 5 litre

CCE159

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Factors Affecting Equilibrium; Le-Chatelier's Principle; Problems on Le-Chatelier's Principle

7. In a closed system : $A(s) \rightleftharpoons 2B(g) + 3C(g)$, if the partial pressure of C is doubled at equilibrium, then partial pressure of B will be

(A) two times the original value
(B) one half of its original value
(C) $\frac{1}{2\sqrt{2}}$ times the original value
(D) $2\sqrt{2}$ times the original value

CCE192

8. $aA \rightleftharpoons bB + cC$, $\Delta H = -x$ kcal.

If high pressure and low temperature are the favourable condition for the formation of the product in above reaction, hence:-

(A) a> b + c
(B) a< b + c
(C) a= b + c
(D) None of them

CCE193

9. The equilbrium constant for the reaction :

$Nz(g) + Oz(g) \rightleftharpoons 2NO(g)$ at 2000 K is 4×10^4. In presence of catalyst the equilibrium is established ten times faster at the same temperature. What is the value of equilibrium constant in presence of catalyst

(A) 40×10^{-4}
(B) 4×10^{-4}
(C) 4×10^4
(D) None

CCE194

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10. In the equilibrium mixture: $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI(g)$, the mole ratio of gases are $\sqrt{2} : \sqrt{2} :10$, respectively. What would be the effect on the mole ratio on adding 5 mole of He gas at constant pressure?

(A) no change
(B) The new molar ratio becomes 1 : 1 : 5
(C) The new molar ratio becomes 2: 2: 5
(D) The new molar ratio becomes $2 : 2 : 5\sqrt{2}$

CCE195

11. For a reaction $H_2 + I_2 \rightleftharpoons 2HI$ at 721 °C the value of equilibrium constant is 50. If 0.5 moles each of H_2 & I_2 is added to the system the value of equilibrium constant will be.

(A) 100
(B) 25
(C) 50
(D) 200

CCE196

12. For a given exothermic reaction, K_p and K_p' are the equilibrium constants at temperatures T_1 and T_2, respectively. Assuming that heat of reaction is constant in temperature range between T_1 and T_2, it is readily observed that ($T_2 > T_1$) :-

(A) $K_p > K_p'$
(B) $K_p < K_p'$
(C) $K_p = K_p'$
(D) $K_p = \frac{1}{K_p'}$

CCE197

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13. Match list –I (equilibrium) with List –II (conditions for reaction) and select the correct answer using the codes given below the lists :-

List–I (Equilibrium)	**List–II (Conditions)**
P. $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$ Endothermic	1. High temperature
Q. $2AB_2(g)+B_2(g) \rightleftharpoons 2AB_3(g)$ Exothermic	2. Low temperature
R. $2AB_3(g) \rightleftharpoons A_2(g) + 3B_2(g)$ Endothermic	3. High pressure 4. Low pressure 5. Independent of pressure

CODE :

	P	**Q**	**R**
(A)	1 & 3	2 & 3	2 & 4
(B)	2 &3	1 & 4	1 & 3
(C)	1 & 5	2 & 3	1 & 4
(D)	2 & 4	1 & 5	1 & 3

CCE157

14. Match the following

Column-I (Reactions)	**Column-II (Effect on addition of water)**
(A) $A(aq) \rightleftharpoons 2B(aq) + H_2O(l)$	(P) Moles of 'A' increases
(B) $3A(aq) \rightleftharpoons 2B(aq)$	(Q) Moles of 'B' increases
(C) $2A(aq) + H_2O(l) \rightleftharpoons B(aq)$	(R) Molar concentration of 'A' decreases
(D) $2A(aq) \rightleftharpoons 3B(aq)$	(S) Molar concentration of 'B' decreases

CCE160

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Equilibrium in Physical Processes

15. At constant temprature on increasing pressure which is correct :

(A) Solid $\rightleftharpoons$ Liquid more liquid formed

(B) Formation of ammonia from N_2 and H_2 decreases

(C) $CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$ reaction will shifted in forward direction

(D) Solubility of gas in liquid increases

CCE198

ANSWER KEY

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Answer	B	A	B	B	A	D	C	A	C	A
Question	**11**	**12**	**13**	**14**						**15**
Answer	C	A	C	A → Q, R, S; B → P, R, S; C → P, R, S; D → Q, R, S						D

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